

Overview of CCTV, NVR, and DVR

Proprietary Video Formats

Modern Closed-Circuit Television (CCTV), Network Video Recorder (NVR), and Digital Video Recorder (DVR) systems export raw video recordings wrapped in proprietary container formats. These custom containers serve critical functions in video surveillance operations, primarily ensuring evidence integrity, tamper proofing, precise frame synchronization, and the inclusion of embedded metadata (such as timestamps, camera identification, and Intelligent Video Surveillance / IVS analytics).

Key Reasons for Proprietary Formats

- **Cryptographic Watermarking:** Guarantees that video footage presented in legal proceedings has not been altered, cut, or edited.
- **Embedded Analytics Metadata:** Retains motion detection boxes, heatmaps, line-crossing events, and audio metadata natively alongside the video track.
- **Frame-Accurate Timestamps:** Synchronizes video playback across multiple channels down to the exact millisecond.

Major CCTV Manufacturers & Proprietary Formats

Manufacturer / Brand	Proprietary Container Extension	Standard Export Formats	Recommended Native Player Software
Hikvision / HiLook / Annke	.dav, .mp4 (custom wrapper), .exe	MP4, AVI	Hikvision VSPlayer
Dahua / Lorex / Amcrest	.dav	MP4, ASF, AVI	Dahua Smart Player
Bosch Security	.mp4 (custom metadata wrapper), .bvip	MP4, AVI	Bosch Video Client / BVMS
Pelco	.pef, .pns, .exp, .xpa	AVI, MP4	Pelco Export Player / DS Media Player
Axis Communications	.asf, .mkv (custom metadata)	MP4, WebM	Axis File Player
Hanwha Vision	.sec	MP4, AVI	Wisenet SSM /

Manufacturer / Brand	Proprietary Container Extension	Standard Export Formats	Recommended Native Player Software
(Samsung)			Backup Player
Swann	.dav, .vob, .h264, .media	MP4, AVI	SwannView / Smart Player
Generic / Unbranded (XMEye / TVT / Avtech)	.264, .265, .avf, .vfv	AVI, MP4	General CCTV Player / CMS Software

Summary of File Extensions

- .dav: The most widely used proprietary CCTV format across Dahua, Hikvision, Lorex, and OEM brands.
- .sec: Native encrypted format used by Hanwha/Samsung Wisenet systems to secure footage.
- .pef / .pns: Proprietary formats for Pelco Endura and Digital Sentry VMS platforms.
- .exe: Self-extracting video standalone executables containing embedded player applications for easy playback without software installation.